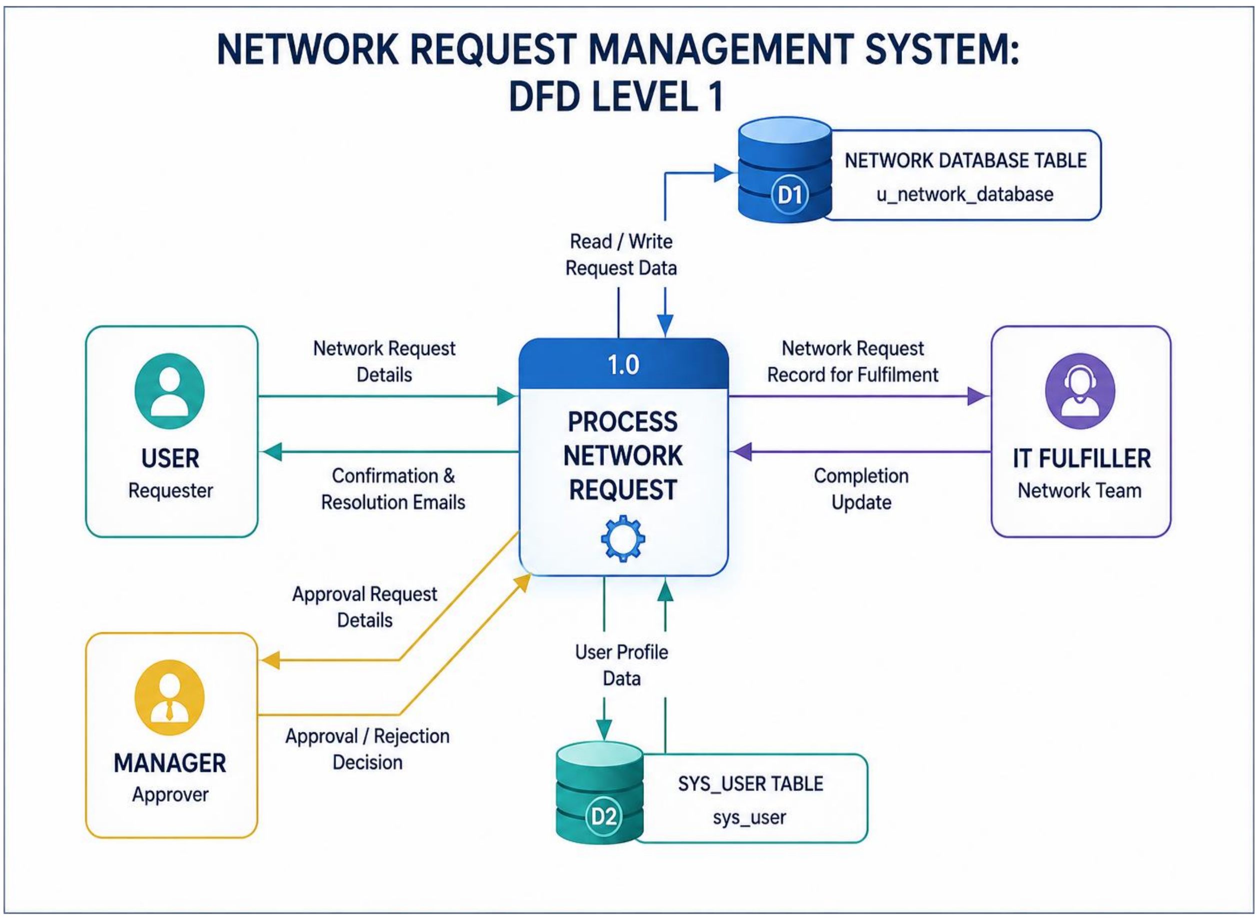
Requirement Analysis

Data Flow Diagram & User Stories

Date	**29-06-2026**
Project Name	**Automated Network Request Management**
Team ID	
Maximum Marks	

Part 1: Data Flow Diagram (DFD)



Simplified Flow Steps:

Phase 1: Request Intake & Data Validation

- **Data Origin:** The end-user submits a request via a customized form on the ServiceNow Service Portal.
- **Input Data:** Network parameters such as Source IP, Destination IP, Target Port, and Business Justification are captured.
- **Processing:** ServiceNow client-side and server-side UI Policies validate and sanitize inputs (e.g., regex validation for IP format, mandatory field checks).
- **Output:** A validated request item (sc_req_item) is created in the system.

Phase 2: Governance & Workflow Orchestration

- **Process Engine:** ServiceNow Flow Designer evaluates the request against predefined business rules.
- **Risk Assessment:** CI relationships from the CMDB are analyzed to determine risk level.
- **Decision Logic:**
    - High-risk requests: Routed for Manager/Security approval (Approved / Rejected decision required).
    - Low-risk requests: Automatically approved and forwarded to integration processing, bypassing manual intervention.

Phase 3: Integration & Infrastructure Provisioning

- **Transformation Layer:** Integration Hub converts ServiceNow data into structured payloads (JSON/XML) suitable for external systems.
- **Outbound Communication:** Secure REST API calls or orchestration via Ansible spokes are triggered.
- **Target Systems:** Network platforms such as Palo Alto Panorama, Cisco DNA Center, or Infoblox IPAM receive the request.
- **Execution:** Network controllers apply the configuration changes on physical or cloud infrastructure**.**

Phase 4: Verification & Audit Logging (Closed Loop)

- **Response Handling:** Infrastructure systems return execution status (e.g., 200 OK, error codes) along with logs**.**
- **Processing:** ServiceNow parses and evaluates the response.
- **Post-Processing:**
  - On success: CMDB is updated with new configuration state and an immutable audit log is generated.
  - On failure: Error is logged and notification is triggered for remediation.
- **Notification:** End-user receives automated confirmation of request status (success/failure)

Part 2: User Stories

Here is a set of comprehensive, production-ready Agile User Stories tailored for the **Automated Network Request Management (ANRM)** ServiceNow project, structured in the requested tabular format.

ServiceNow ANRM User Stories & Tasks

User Type	Functional Requirement	User Story / Task	Acceptance Criteria	Priority
End-User *(Developer / Requestor)*	**Self-Service Request Intake**	**As an** End-User,**I want** a dedicated Service Catalog form for firewall port requests,**So that** I can request access without tracking down a network engineer.	• Form must include mandatory fields: Source IP, Destination IP, Ports, Protocol (TCP/UDP), and Business Justification. • Form must dynamically show/hide advanced routing choices based on user role.	**High**

IT Ops Manager *(Approver)*	**Automated Governance / Approval Routing**	**As an** IT Operations Manager,**I want** low-risk requests auto-approved and high-risk requests routed to my queue,**So that** standard requests aren't delayed by manual approvals.	• Flow Designer must evaluate request criteria against a predefined risk matrix. • Requests with known standard ports (e.g., 443) auto-approve instantly. • High-risk ports (e.g., 22, 3389) trigger an email/system approval notification to the manager.	**High**

ServiceNow Dev *(Fulfiller)*	**Integration Hub API Integration**	**As a** ServiceNow Developer,**I want** Integration Hub to trigger REST API payloads to the network controller,**So that** configuration changes are applied programmatically.	• Upon approval token, Integration Hub must compile form data into a valid JSON payload. • Payload must be securely sent to the target API endpoint (e.g., Palo Alto/Ansible) using encrypted credentials. • System must capture synchronous HTTP response codes (200 OK for success).	**Critical**

Network Administrator *(Auditor)*	**Automated Verification & CMDB Logging**	**As a** Network Admin,**I want** the system to test the connection and log details to the CMDB,**So that** our inventory remains accurate and compliant.	• Post-execution step must verify connectivity status. • On success, the script must write a configuration relationship record back to the target CI in the CMDB. • The RITM (Requested Item) ticket must lock down an uneditable audit log containing who requested, who approved, and API logs.	**Medium**

End-User *(Developer / Requestor)*	**Real-Time Notification & Visibility**	**As an** End-User,**I want** real-time status updates on my request,**So that** I don't have to send follow-up emails to see if it's done.	• System must trigger automated notifications (Email or Slack/Teams webhook) at three milestones: Submitted, Approved/In-Flight, and Fulfilled/Closed. • If an API push fails, notification must alert NetOps with the failure logs.	**Low**